TOYA TAKAHASHI

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EDUCATION

Massachusetts Institute of Technology (M.I.T.) May 2026
B.S. in Electrical Engineering and Computer Science — GPA: 5.0/5.0

EXPERIENCE

NVIDIA Isaac for Manipulation August 2026 - Present
Systems Software Robotics Solutions Engineer *Santa Clara, CA*

- Developing an imitation-learned policy for a robotic manipulator to perform partner-defined tasks, targeting a 95%+ success rate.
- Designing a generalizable AI agent-driven workflow for building an end-to-end robotic manipulation software stack.

MIT Sensing, Perception, Autonomy, and Robot Kinetics (SPARK) Lab September 2024 - May 2026
Undergraduate Researcher *Cambridge, MA*

- Integrated learning-based keypoint extraction with SuperGlue and LightGlue into a visual loop-closure module for SLAM.
- Built a software platform for mobile robots supporting RTK positioning and LiDAR-inertial odometry to generate maps using 3D scene graphs.
- Evaluated camera-LiDAR calibration algorithms to reduce reprojection errors and improve 3D scene mapping accuracy.

NVIDIA Isaac ROS May 2025 - August 2025
Systems Software Robotics Solutions Intern *Santa Clara, CA*

- Developed a CUDA-accelerated suite of ROS 2 nodes for running a TensorRT-optimized Grounding DINO open-set object detection model, including a text tokenizer, text and image tensor encoders, and an output tensor decoder with visualization.
- Integrated Segment Anything (SAM, SAM2) and Grounding DINO into a robotic manipulator workflow using behavior trees to segment and detect arbitrary objects for perception-driven pick-and-place tasks.
- Implemented a C++ ROS node to align Realsense depth images from the IR to RGB frame, enabling deep-learning-enhanced stereo disparity matching for accurate pose estimation.

MIT Arcturus Robotics September 2022 - March 2025
Autonomy Software Team Co-Lead *Cambridge, MA*

- Led a software team of approximately 20 students in developing an Autonomous Surface Vehicle (ASV) autonomy stack using C++ and Python with ROS 2 and Mission Oriented Operating Suite (MOOS-IvP) middlewares.
- Developed an algorithm to overlay clustered LiDAR point cloud onto the camera frame to match obstacles with detected objects.
- Implemented an Extended Kalman Filter to fuse GPS and IMU data for global robot localization with centimeter-level accuracy.

NVIDIA Isaac ROS May 2024 - August 2024
Systems Software Robotics Solutions Intern *Santa Clara, CA*

- Enhanced an end-to-end robot manipulator object-following workflow by improving stability of object pose estimations from a deep neural network.
- Implemented ROS nodes for post-processing pose streams via averaging, stability analysis, outlier detection, and Kalman filtering.
- Developed and optimized a CUDA-accelerated alpha compositing ROS node, enabling efficient GPU-based image blending without redundant CPU-GPU memory transfers.

TECHNICAL SKILLS

Computer Languages	Python, C/C++, Go, NumPy, CUDA, SystemVerilog, RISC-V Assembly
Tools	Git, Docker, Linux, Robot Operating System 2 (ROS 2), Nix, PyTorch, Drake